Stockholm University

Department of Meteorology

Modelling of Large-Scale Circulation in Atmosphere and Ocean (MO8004)

Ocean Gyre Circulation
Sverdrup and Munk Solutions

June 3, 2026

Author: Fredrik Bergelv (fredrik.bergelv@live.se)

Supervisor: Hanna Winge

1 Introduction

Due to the Hadley cells together with the Coriolis effect, tradewinds and westerlies dominate the winds over the North Atlantic [Hartmann, 2016]. These winds drive a converging Ekman transport, which is stronger in the North due to the lateral dependence on the Coriolis force [Ekman, 1905]. This imbalance is known as the Sverdrup circulation [Sverdrup, 1947], and dominates most extratropical oceans.

Stommel [1948] further developed the solution, by introducing drag which gave return flow on the western boundary. The western boundary current was later further developed by Munk [1950], who assumed a viscous fluid. In this report, the Sverdrup and Munk solutions will be examined by using a set of shallow water equations with a Fortran code. This will give insight into the wo ocean gyre theories and how a numerical model handles their approximations.

2 Theory

One can approximate the wind stress in the North Atlantic with

$$\tau = \rho_{\text{air}} C_D U_{10}^2 \sin\left(\frac{\pi y}{L_y}\right), \tag{1}$$

where ρ_{air} is the air density, C_D is the drag coefficient, U_{10} is the ten meter wind speed, and L_y is the lateral extent of the basin. The sinusoidal dependence represents easterly trade winds in the southern part of the North Atlantic and westerlies in the northern part.

2.1 Sverdrup Theory

In order to study Sverdrup circulation one needs to make assumptions: One needs a set of linear and stationary shallow water equations, a meridionally varying ocean surface stress, the beta plane approximation as

$$f \approx f_0 + \beta y, \tag{2}$$

and an inviscid fluid on a flat bottom. One also needs the ocean to be laterally bounded, for example by two continents on either side [Vallis, 2006]. One also assumes a barotropic ocean. This is however not completely true, since we also assume an Ekman layer with Ekman transport, but since the size of this layer is much smaller than the domain, it is sufficient to say that the ocean is barotropic. The barotropic streamfunction is defined as

$$\frac{\partial \Psi}{\partial x} = vH, \quad \frac{\partial \Psi}{\partial y} = -uH, \tag{3}$$

where H is the mean depth of the ocean. Using this one obtains the Sverdrup balance as

$$\beta v H = \frac{1}{\rho} \nabla \times \vec{\tau}. \tag{4}$$

One can thus derive the Sverdrup solution, by assuming $\Psi(x_{\text{East}} y) = 0$, as

$$\int_x^{x_{\text{East}}} \frac{\partial \Psi}{\partial x}\, dx = \int_x^{x_{\text{East}}} vH\, dx = \cancel{\Psi(x_{\text{East}}, y)} - \Psi(x, y) \tag{5}$$

$$\implies \Psi(x, y) = -\frac{1}{\rho\beta} \int_x^{x_{\text{East}}} (\nabla \times \vec{\tau})\, dx, \tag{6}$$

The Sverdrup solution thus gives a net southward flow in the center of the domain. We will thus see a "parabolic-like" structure along the longitudinal direction, with the decreasing values towards the East. The western boundary will however have zero flux, and the streamlines will not be closed.

2.2 Munk Theory

In traditional Munk theory one also add meridional harmonic Newtonian viscosity to the shallow water equations. The viscosity is diffusive, which agrees with the fact that the ocean is turbulent and has an effective viscosity [Vallis, 2006]. The reason for only assuming meridional viscosity, is that this was the only net velocity in the Sverdrup solution. By assuming that the solution is bounded by the domain, $\Psi(x_{\text{East}}) = \Psi(x_{\text{West}}) = 0$, one obtains

$$A\nabla^4\Psi - \beta\frac{\partial\Psi}{\partial x} = \frac{1}{\rho}\nabla\times\vec{\tau} \implies \tag{7}$$

$$\Psi(x,y) = \Psi_{\text{Sverdrup}}(x,y)\left(1 - e^{-\frac{x-x_{\text{West}}}{2\delta_m}}\left[\cos\left(\sqrt{3}\frac{x-x_{\text{West}}}{2\delta_m}\right) + \frac{1}{\sqrt{3}}\sin\left(\sqrt{3}\frac{x-x_{\text{West}}}{2\delta_m}\right)\right]\right), \tag{8}$$

where δ_M is the Munk layer defined as

$$\delta_M \equiv \left(\frac{A}{\beta}\right)^{\frac{1}{3}}. \tag{9}$$

The Munk layer is the distance from the boundary to the latitude of largest northward meridional velocity. One thus assumes that the interior should behave as the Sverdrup solution. Due to the non-linearity of the Munk theory will also affect the shape of the theoretical solution. This will yield closed streamlines at the western boundary, with a western boundary current at the distance δ_M from the western boundary. However the Sverdrup interior solution will remain.

3 Methodology

A model written in Fortran was used that implemented the linear shallow water equations, with a imposed stress described by Equation 1. Otherwise, the code was initialized at rest and used an Euler step for the first step, then a leapfrog scheme on a Arakawa C-grid as presented by Döös et al. [2022]. The code ran for at least 14 days, and was saved for every 12th hour. The code can be observed in Appendix A.

cournot number

The code used a domain depth of 3000 m, $f_0 = 7.27\times10^{-5}\,\mathrm{s}^{-1}$ and $\beta = 1.98\times10^{-11}\,\mathrm{m}^{-1}\,\mathrm{s}^{-1}$. The model used a domain with solid boundaries of the size $L_x = L_y = 7000\,\mathrm{km}$ (dubbed the width for the Pacific Ocean) with $\Delta x = \Delta y = 35\,\mathrm{km}$. VArying the stress was done by using $U_{10} = [4\,\mathrm{m\,s^{-1}}, 5\,\mathrm{m\,s^{-1}}, 6\,\mathrm{m\,s^{-1}}]$, and varying the viscosity was done by $A = [3\times10^4\,\mathrm{m^2\,s^{-1}}, 1\times10^5\,\mathrm{m^2\,s^{-1}}, 5\times10^5\,\mathrm{m^2\,s^{-1}}]$. The harmonic viscosity in the boundary points, assumed zero flow outside the domain.

4 Results

4.1 Sverdrup circulation

The Sverdrup solution gave Figure 1–3. One can in Figure 1 see the result of the f-plane, which does not resemble the beta Sverdrup theory. Here the solution is symmetric, since there is no latitudinal asymmetry caused by the β-term. Furthermore, the solution is growing with time, due to the continuous input of wind energy. This is a growing geostrophic solution to the applied wind stress, and thus symmetric in the domain. However, an gyre can still be seen due to the consequence of the y-dependence on the wind stress.

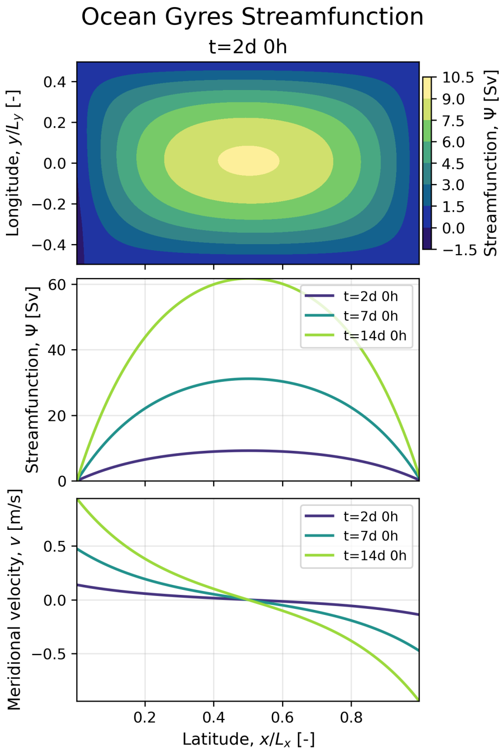

Figure 1: Streamfunction for the no-viscosity f-plane configuration. The circulation remains symmetric with no β-effect and without western intensification.

The introduction of the β-term gave Figure 2. One can observe Sverdrup like solution compared to Figure 1, due to the asymmetric beta effect (stronger f northward and weaker southward) pushing the solution towards the western boundary. The mirror-symmetry is thus broken and a better Sverdrup solution is obtained.

Varying the wind stress gave Figure 2. There one can see that a stronger wind linearly increases the flux in the domain. This is reasonable since the wind stress is the forcing that drives this circulation. Thus, more forcing yields larger fluxes. Examining the slope, one sees larger southward fluxes in the interior. Overall, one can see okay agreement for the time-averaged result and the theory. Away from the western boundary the model and the theory align very good, but close to the western boundary they are very bad.

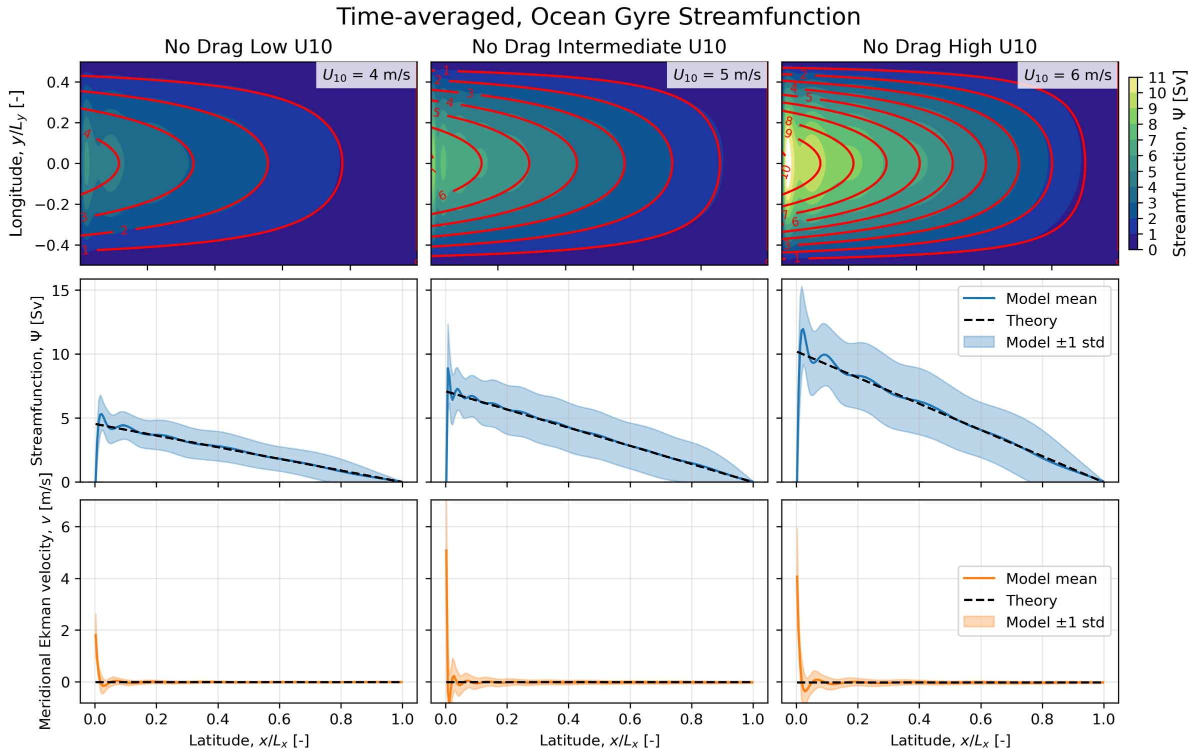

Figure 2: Comparison of streamfunction fields for different ten meter wind speeds. Increasing wind speeds shows that it increases the magnitude of the wind-driven ocean gyre. The contours show the model result, while the solid red lines showcase the Sverdrup theory. The cross-sections of the streamfunction is at the center of the domain. The meridional wind velocity has been transformed into an Ekman-layer velocity, by assuming an Ekman depth of $\delta_{\mathrm{EK}} = 50\,\mathrm{m}$. The meridional velocity is thus the corresponding mass flux as if the domain is at rest and only the Ekman layer is moving. The time averaging was done between two days and two weeks for the model.

In Figure 3 one can observe the effect if one increases the domain width, while keeping Δx constant. One can observe that a larger flux for the North Pacific compared to the North Atlantic, although the meridional velocity is exactly the same in the two oceans (as seen from the slope of the streamfunction). However, the error remains the same.

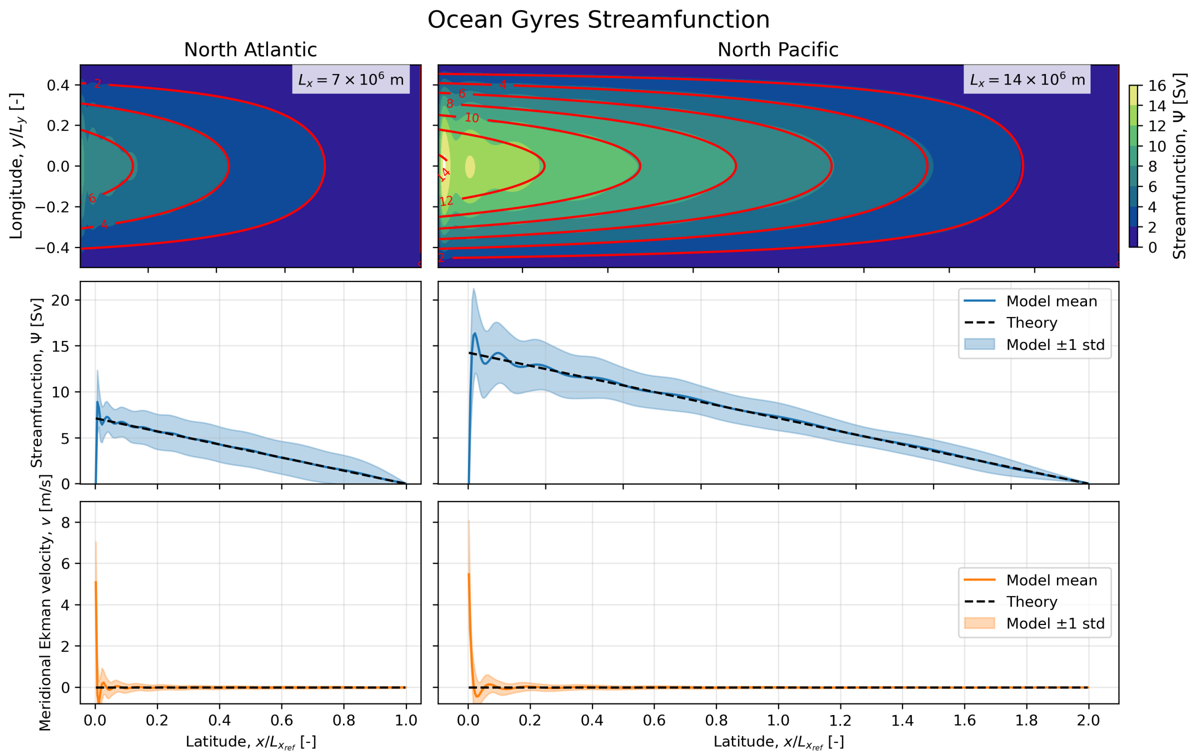

Figure 3: Streamfunction for the no-viscosity β-plane configuration with varying domain size. The contours show the model result, while the solid red lines showcase the Sverdrup theory. In the center domain cross-section, solid lines show the numerical model results, while dashed lines show Sverdrup theory. The meridional wind velocity has been transformed into an Ekman-layer velocity, by assuming an Ekman depth of $\delta_{\mathrm{EK}} = 50\,\mathrm{m}$. The meridional velocity is thus the corresponding mass flux as if the domain is at rest and only the Ekman layer is moving. The time averaging was done between two days and two weeks for the model.

4.2 Munk's theory

The difference of Munk viscosity gave Figure 4. One can observe that the Munk theoretical results now include the western boundary current, in both the numerical and theoretical solution. Thus, the numerical solutions agrees better with the theoretical, compared to the Sverdrup solution. The largest distinction is the steady state convergence of the Munk viscosity, which does not occur for the Sverdrup solution. Observing the instantaneous relative error one can see a convergence towards lower errors for the Munk solution, while the same cannot be said for the Sverdrup solution. One thus needs to time average the Sverdrup result to obtain good values. One can observe that the time-averaged Sverdrup solution gives similar accuracy to the instantaneous Munk solution. However, to make a fair compression between the two models, the Munk solution was also time averaged.

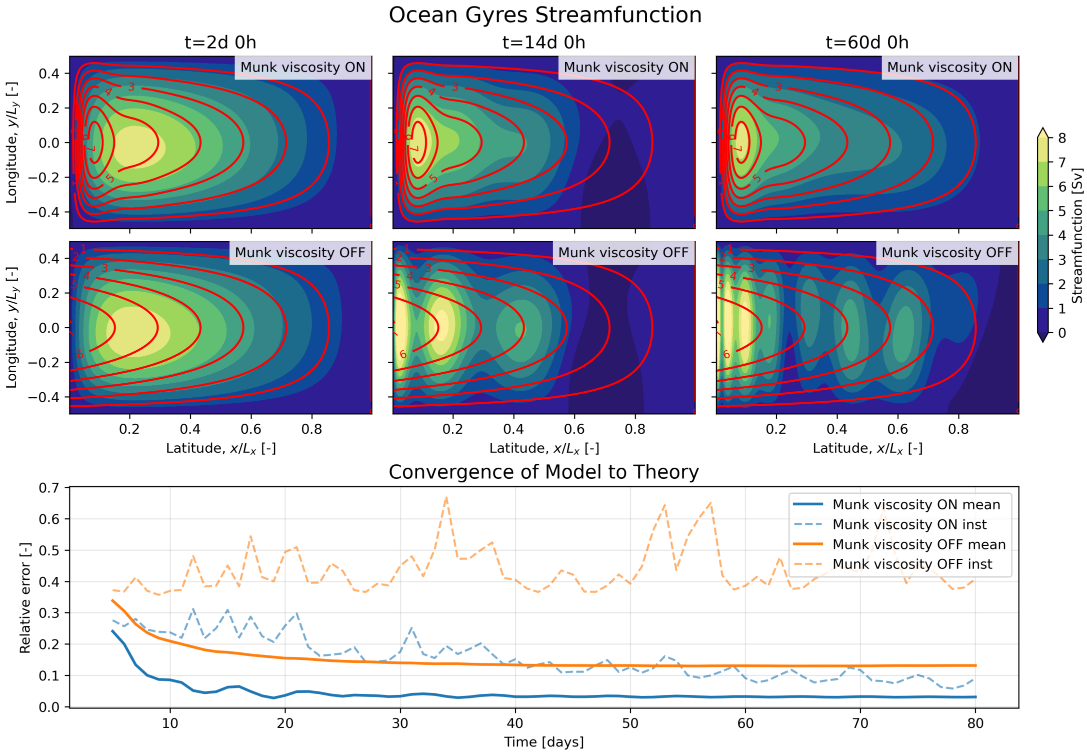

Figure 4: Streamfunction for the Laplace-viscous β-plane configuration. This figure shows the effect of the Munk friction, compared to the Sverdrup theory without friction. The red lines are the theoretical Sverdrup and Munk solutions respectively.

Munk theory gave Figure 5. Increasing the viscosity coefficient A can be seen to do two major things: increasing the speed of the western boundary current and moving the western boundary current further away from the western boundary. The western boundary current is moved away from the boundary since this thickens the Munk layer δ_M, which can be observed in the figure. Compared to the Sverdrup solution one observes better result, especially for the western boundary current.

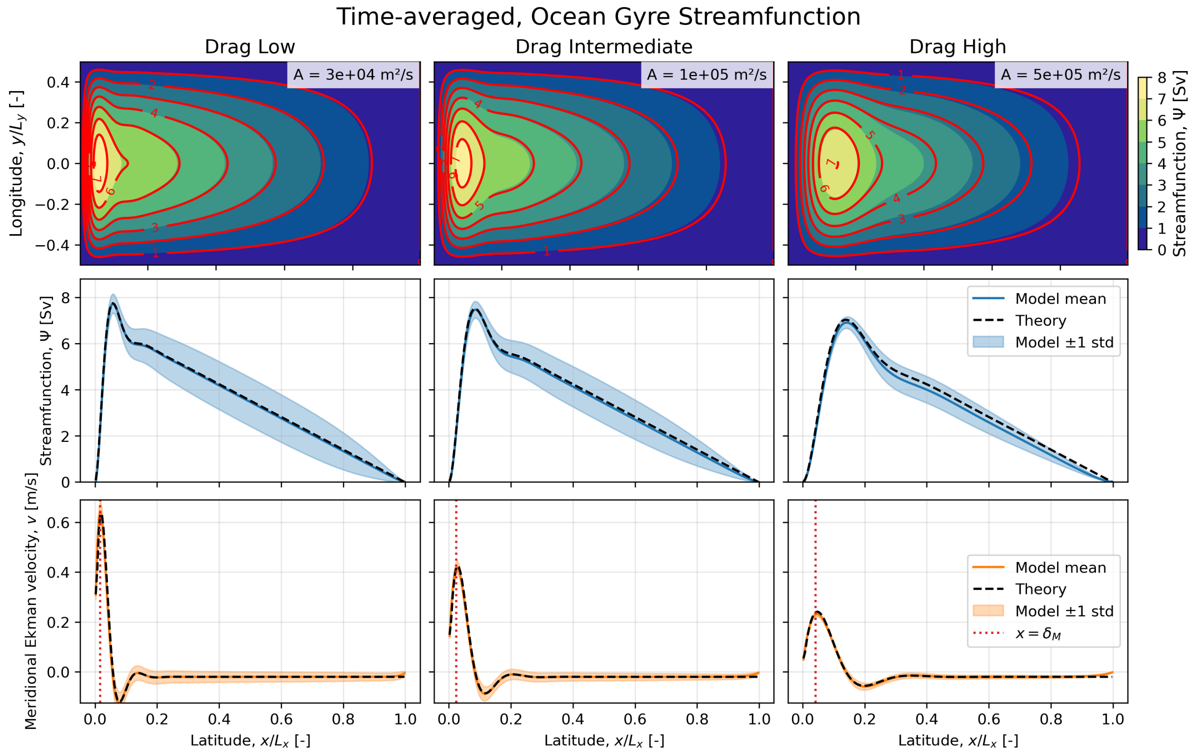

Figure 5: Streamfunction fields for different values of the meridional viscosity coefficient A, compared to Munk theory. The contours show the model result, while the solid red lines showcases the theory. The resolution of the Munk depth is $\frac{\delta_M}{\Delta x} = [3.8,\ 4.9,\ 8.4]$, from left to right in the figure. The center domain zonal cross-sections show solid lines for the numerical model, while dashed lines represent Munk theory. The meridional wind velocity has again been transformed into an Ekman-layer velocity, by assuming an Ekman depth of $\delta_{\text{EK}} = 50\,\text{m}$. The meridional velocity is thus the corresponding mass flux as if the domain is at rest and only the Ekman layer is moving. The time-averaging was done between two days and two weeks.

5 Discussion

The reason for the western intensification can be explained by the phase movement of Rossby waves always being westward. Thus larger fluxes appear on the western half. In the Munk theory, the imposed harmonic viscosity ensured that these Rossby waves died down, ensuring a stable solution. However, the lack of diffusion in the Sverdrup solution forced continuous Rossby wave propagation, which explains the non-converging result. Since the Rossby wave propagation was superimposed on the solution, time averaging resulted in good results.

This non-convergence can be explained through the lack of energy dissipation. Through continuous energy input, propagation of Rossby waves was continuously fed. This is the same mechanism that forced the f-plane solution to grow continuously. Thus one needs some kind of dissipation to balance the continuous energy input. In the real world this is done through small-scale dissipation, connections to other oceans and other dynamical properties.

Another problem of the Sverdrup solution is the lack of mass convergence in the theory. Since the theory is linear, only one boundary condition could be imposed, which cannot allow return flow. The Munk solution is non-linear, allowing multiple boundary conditions and a return flow. The return flow can be explained through the frictional effect opposing the mean southward flow, by creating a boundary layer close to the western boundary. Since frictional effects act in the opposite direction, a strong northern current is created.

Overall, clear limitations of the Sverdrup model can be observed due to the accumulation of mass along the western boundary. This leads to the formation of a strong western boundary current, which is not represented in the original Sverdrup theory. Therefore, some form of viscosity or friction must be included in the system to properly balance the boundary transport. Another clear limitation is the fact that the solution does not reach steady state (as seen in Figure 5), which forces one to do a time-averaging to get realistic results.

One can furthermore criticize the Munk model for artificial viscosity. If the Stommel drag is the friction caused by bottom drag, the Munk viscosity should be a boundary effect, and thus only be seen at the boundaries. However, since the viscosity presents reasonable results, one can argue that the model is of good use. However, in most realistic models one often uses the combined Munk–Stommel viscosity to obtain benefits of both models [Vallis, 2006].

One can always obtain better results by increasing the grid size in the domain; however, according to Steele [2008], the speed of the Gulf Stream is around $2.5\,\mathrm{m\,s^{-1}}$. Comparing with Figure 2–5, the obtained results in these schemes are thus on the right order of magnitude compared to the real world. Thus, even though this article presented a simplified model, reasonable values were obtained. Furthermore, the error remained below 15% for both models, showcasing working numerical models.

6 Conclusion

In conclusion, both models have their strengths and weaknesses respectively. Sverdrup theory is a good simple way to visualize the flow in the upper ocean, while Munk theory is good if one also wants to resolve the western boundary current. Three important take-aways are:

- Both numerical models showed western boundary intensification, explained by Rossby wave propagation. However these Rossby waves must be balanced by some kind of friction to not grow with time and cause non-converging behaviour.
- Although both numerical models showed return flow, only Munk theory showed a western boundary current, explained by a frictional layer.
- A larger domain gave the same interior meridional velocity, which resulted in a stronger western boundary current due to the interior larger flux of the larger ocean.

One can summarize the two models by stating that although the Sverdrup theory is more realistic from a physical perspective, the artificial Munk theory presented a more realistic solution. This showcases a scenario where the physical theory and the actual models do not align. However, a good physicist would then know when to use one model or the other, to explain different aspects of the ocean gyre circulation.

References

Kristofer Döös, Peter Lundberg, and Aitor Aldama Campino. *Basic Numerical Methods in Meteorology and Oceanography.* Stockholm University Press, Stockholm, Sweden, 2022. ISBN 978-91-7635-172-7. doi: 10.16993/bbs.

V. Walfrid Ekman. On the Influence of the Earth's Rotation on Ocean-Currents. *Arkiv för matematik, astronomi och fysik*, 2(11):1–52, May 1905.

Dennis L. Hartmann. *Global Physical Climatology.* Elsevier, second edition edition, 2016. ISBN 978-0-12-328531-7.

Walter H. Munk. On the wind-driven ocean circulation. *Journal of Meteorology*, 7(2):79–93, 1950.

John H. Steele. *Encyclopedia of Ocean Sciences.* Academic Press, Amsterdam, 2 edition, 2008. ISBN 9780123744739.

Henry Stommel. The westward intensification of wind-driven ocean currents. *Transactions, American Geophysical Union*, 29(2):202–206, 1948.

H. U. Sverdrup. Period increase of ocean swell. *Eos, Transactions American Geophysical Union*, 28(3): 407–417, 1947.

G. K. Vallis. *Atmospheric and Oceanic Fluid Dynamics.* Cambridge University Press, Cambridge, U.K., 2006.

A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/`
`Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`